Package ‘tisai’

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Title Datasets and Utilities for AI-Enabled Time Series and Spatial Statistics

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Description Provides datasets and helper functions used in the book
``AI Enabled Time Series and Spatial Statistics''.

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Imports dplyr,
quantmod,
zoo

LazyData true

Contents

`AAPL` *Daily Apple Stock Market Data*

Description

Daily Apple Inc. stock prices, trading volume, cash dividends, and split events from 2010 to 2026.

Usage

```
data(AAPL)
```

Format

A data frame with 4000 observations on the following 8 variables:

`date` Trading date from April 23, 2010 to March 18, 2026.

`open` Opening price in U.S. dollars.

`high` Highest price of the day in U.S. dollars.

`low` Lowest price of the day in U.S. dollars.

`close` Closing price in U.S. dollars.

`volume` Trading volume.

`dividend` Cash dividend per share on dividend dates, and 0 otherwise.

`split` Split factor on stock split dates, and 0 otherwise.

Details

This dataset is drawn from the Kaggle collection “Apple AAPL Full Stock Financial Dataset 1980–2026”. Following the user-approved trimming rule for this project, the package version keeps the latest 4000 trading-day observations, which currently span April 23, 2010 to March 18, 2026. The resulting data frame combines daily prices, trading volume, dividend records, and split events for Apple Inc. in a single object.

Source

Anadi Gupta, "Apple AAPL Full Stock Financial Dataset 1980–2026", Kaggle. https://www.kaggle.com/datasets/anadiskt/apple-aapl-full-stock-financial-dataset-1980-2026

The dataset description reports underlying data sources including Yahoo Finance, Nasdaq, and public financial disclosures.

Examples

```
data(AAPL)
head(AAPL)

plot(close ~ date, data = AAPL, type = "l",
     xlab = "Date", ylab = "USD",
     main = "Apple Closing Price")
```

`Ashare` *Selected Chinese A-Share Stock Tickers*

Description

A list of representative Chinese A-share listed companies used for constructing panel stock return data from Yahoo Finance.

Usage

```
data(Ashare)
```

Format

A data frame containing stock tickers of selected A-share companies:

`ticker` Yahoo Finance ticker symbol.

`name` Company name.

`market` Exchange code: SSE or SZSE.

Source

Yahoo Finance symbol convention for Chinese A-share listed companies: https://finance.yahoo.com/

Examples

```
## Not run:
data(Ashare)

## download panel data for selected A-share stocks
panel <- get_Ashare_data(
  tickers = Ashare$ticker[1:5],
  from = "2020-01-01",
  to = "2024-12-31"
)
```

```
head(panel)

## End(Not run)
```

CNgdp *China Quarterly GDP and Sectoral Value Added*

Description

Quarterly measures of China's gross domestic product (GDP), including both current-price (nominal) and constant-price (real) series, together with value added of the primary, secondary, and tertiary industries.

Usage

```
data(CNgdp)
```

Format

A data frame with quarterly observations on:

`date` First day of the quarter (YYYY-MM-DD).

`gdp` GDP, current price (billion CNY).

`gdp1` Primary industry value added, current price (billion CNY).

`gdp2` Secondary industry value added, current price (billion CNY).

`gdp3` Tertiary industry value added, current price (billion CNY).

`rgdp` GDP, constant price (billion CNY).

`rgdp1` Primary industry value added, constant price (billion CNY).

`rgdp2` Secondary industry value added, constant price (billion CNY).

`rgdp3` Tertiary industry value added, constant price (billion CNY).

Source

National Bureau of Statistics of China (NBS), Quarterly Data Portal: https://data.stats.gov.cn/

`CNpcgdp` *Provincial Per Capita GDP in China, 2016–2024*

Description

Annual per capita GDP for 31 provincial-level regions in China from 2016 to 2024.

Usage

```
data(CNpcgdp)
```

Format

A data frame with 31 observations on the following 10 variables:

`province` Provincial-level region name.

`pcgdp2016` Per capita GDP in 2016 (CNY per person).

`pcgdp2017` Per capita GDP in 2017 (CNY per person).

`pcgdp2018` Per capita GDP in 2018 (CNY per person).

`pcgdp2019` Per capita GDP in 2019 (CNY per person).

`pcgdp2020` Per capita GDP in 2020 (CNY per person).

`pcgdp2021` Per capita GDP in 2021 (CNY per person).

`pcgdp2022` Per capita GDP in 2022 (CNY per person).

`pcgdp2023` Per capita GDP in 2023 (CNY per person).

`pcgdp2024` Per capita GDP in 2024 (CNY per person).

Source

National Bureau of Statistics of China (NBS), data portal: https://data.stats.gov.cn/

`CNpower` *China Monthly Electricity Generation by Source*

Description

Monthly electricity generation in China, including total generation and generation by thermal, hydro, nuclear, wind, and solar sources.

Usage

```
data(CNpower)
```

Format

A data frame with monthly observations on:

`date` First day of the month (YYYY-MM-DD).

`power` Total electricity generation (100 million kWh).

`thermal` Thermal power generation (100 million kWh).

`hydro` Hydropower generation (100 million kWh).

`nuclear` Nuclear power generation (100 million kWh).

`wind` Wind power generation (100 million kWh).

`solar` Solar power generation (100 million kWh).

Source

National Bureau of Statistics of China (NBS), Monthly Data Portal: https://data.stats.gov.cn/

`CNprice` *China Monthly Consumer Price Indices (YoY)*

Description

Monthly consumer price indices (CPI), reported as year-over-year indices (same month of previous year = 100), for the national aggregate as well as urban and rural residents.

Usage

```
data(CNprice)
```

Format

A data frame with monthly observations on:

`date` First day of the month (YYYY-MM-DD).

`cpi` National CPI (YoY index, same month last year = 100).

`cpi_city` Urban CPI (YoY index, same month last year = 100).

`cpi_rural` Rural CPI (YoY index, same month last year = 100).

Source

National Bureau of Statistics of China (NBS), Monthly Data Portal: https://data.stats.gov.cn/

DelhiWeather *Daily Weather and Air Quality Data for Selected Delhi Sites*

Description

Daily weather and air-quality summaries for six monitoring locations in Delhi, India, during 2025.

Usage

```
data(DelhiWeather)
```

Format

A data frame with 2190 observations on the following 15 variables:

`location` Monitoring location in Delhi.

`date` Observation date in 2025.

`lat` Latitude of the monitoring location.

`lon` Longitude of the monitoring location.

`temperature` Daily mean temperature in degrees Celsius.

`humidity` Daily mean relative humidity percentage.

`pressure` Daily mean atmospheric pressure in millibars.

`windspeed` Daily mean wind speed in kilometers per hour.

`condition` Most frequent hourly weather condition recorded on that day.

`aqi` Daily mean air quality index.

`pm25` Daily mean PM2.5 concentration in micrograms per cubic meter.

`pm10` Daily mean PM10 concentration in micrograms per cubic meter.

`co` Daily mean carbon monoxide concentration in micrograms per cubic meter.

`no2` Daily mean nitrogen dioxide concentration in micrograms per cubic meter.

`obs_hours` Number of hourly observations used in the daily summary.

Details

The original file contains 52560 hourly observations for six Delhi monitoring sites from January 1, 2025 to December 31, 2025. For teaching and package use, the hourly records are aggregated to a daily location-level panel by averaging the numeric indicators and retaining the most frequent hourly weather condition for each site-day.

Source

Sohail Shaikh, “Delhi Weather and AQI Dataset (2025)”, Kaggle. https://www.kaggle.com/datasets/sohails07/delhi-weather-and-aqi-dataset-2025

According to the dataset description, the hourly records were aggregated from real-time weather and air-quality monitoring APIs for the Delhi NCR. The collector code is available at https://github.com/Sohail-Shaikh-07/AI-Agents/tree/main/weather-ai-agent.

Examples

```
data(DelhiWeather)
head(DelhiWeather)

plot(aqi ~ date,
     data = subset(DelhiWeather, location == "Anand Vihar"),
     type = "l",
     xlab = "Date",
     ylab = "Daily mean AQI")
```

---

djia *Daily Dow Jones Industrial Average Prices*

Description

Daily Dow Jones Industrial Average trading data redistributed from the **astsa** package and reorganized for use in **tisai**.

Usage

```
data(djia)
```

Format

A data frame with 2518 observations on the following 6 variables:

`date` Trading date from April 20, 2006 to April 20, 2016.

`open` Opening index level.

`high` Highest index level of the trading day.

`low` Lowest index level of the trading day.

`close` Closing index level.

`volume` Trading volume.

Details

The original **astsa** dataset stores these values as an `xts` object. In **tisai**, the data are reorganized as a base data frame with an explicit `date` column so they can be used more easily in teaching examples and package demonstrations.

Source

Redistributed from the **astsa** package. https://cran.r-project.org/package=astsa https://cran.r-project.org/web/packages/astsa/astsa.pdf

The original **astsa** documentation notes that the market data were obtained from Yahoo Finance via R financial-data tools.

Examples

```
data(djia)
head(djia)

plot(djia$date, djia$close, type = "l",
     xlab = "Date", ylab = "Index level",
     main = "DJIA Close")
```

---

employ — *Annual Global Employment and Unemployment Rates*

Description

Annual country-level employment and unemployment rates for the total population aged 15 and above from 2016 to 2025.

Usage

```
data(employ)
```

Format

A data frame with 1816 observations on the following 5 variables:

- `iso_code` Three-letter country or territory code used by the source database.
- `country` Country or territory name.
- `year` Calendar year from 2016 to 2025.
- `employ` Employment rate, measured as the percentage of employed persons in the working-age population.
- `unemp` Unemployment rate, measured as the percentage of unemployed persons in the labour force.

Details

This dataset is drawn from a Kaggle collection based on ILOSTAT modelled estimates published by the International Labour Organization. The package version is organized as a country-year panel that combines employment and unemployment rates using exact matches on `iso_code`, `country`, `sex`, `age`, and `year`. It keeps the total population aged 15 and above from 2016 to 2025, yielding 1816 observations for 183 countries and territories.

The employment and unemployment rates in this dataset should not be interpreted as complements that add up to 100, because they use different denominators in the source definitions.

Source

Luca Lullo, "Global Employment and Unemployment Rates 1991–2025", Kaggle. https://www.kaggle.com/datasets/lucalullo/global-employment-unemployment-rates-1991-2025

Underlying data source: International Labour Organization (ILO), ILOSTAT database. https://ilostat.ilo.org/

Examples

```
data(employ)
head(employ)

china <- subset(employ, country == "China")
plot(china$year, china$employ, type = "b",
     ylim = range(c(china$employ, china$unemp)),
     xlab = "Year", ylab = "Percent",
     main = "Employment and Unemployment Rates in China")
lines(china$year, china$unemp, type = "b", col = "red")
legend("topright", legend = c("Employment", "Unemployment"),
       col = c("black", "red"), lty = 1, pch = 1, bty = "n")
```

`Energy` *Selected Global Energy Indicators*

Description

Annual energy-related indicators for selected countries.

Usage

```
data(Energy)
```

Format

A data frame containing:

`country` Country name.

`year` Year.

`pop` Population.

`gdp` Gross domestic product.

`energy` Primary energy consumption.

`elec` Electricity generation.

`renew` Share of renewable energy.

Source

Our World in Data energy dataset. https://github.com/owid/energy-data

`EnergyWx` *Daily Spanish Energy Indicators with Weather for Five Major Cities*

Description

Daily Spanish electricity demand, generation, and price indicators combined with city-level weather summaries for five major Spanish cities during 2017 and 2018.

Usage

```
data(EnergyWx)
```

Format

A data frame with 3650 observations on the following 19 variables:

`city` City name: Barcelona, Bilbao, Madrid, Seville, or Valencia.

`date` Observation date from January 1, 2017 to December 31, 2018.

`load` Daily mean actual total electricity load in megawatts.

`loadf` Daily mean forecast total electricity load in megawatts.

`price` Daily mean actual electricity price in euros per megawatt-hour.

`priceda` Daily mean day-ahead electricity price in euros per megawatt-hour.

`gas` Daily mean electricity generation from fossil gas in megawatts.

`hydro` Daily mean hydro reservoir generation in megawatts.

`nuclear` Daily mean nuclear generation in megawatts.

`solar` Daily mean solar generation in megawatts.

`wind` Daily mean onshore wind generation in megawatts.

`temp` Daily mean temperature in degrees Celsius.

`humidity` Daily mean relative humidity percentage.

`pressure` Daily mean atmospheric pressure in hectopascals.

`windspeed` Daily mean wind speed in meters per second.

`rain` Daily accumulated hourly rainfall in millimeters.

`clouds` Daily mean cloud-cover percentage.

`weather` Most frequent hourly weather category recorded on that day.

`obs_hours` Number of distinct hourly weather observations used in the daily summary.

Details

The raw data contain four years of hourly Spanish electricity variables and hourly weather observations for five cities. To satisfy the project size rule while retaining all five cities, this package version keeps the user-approved period from January 1, 2017 to December 31, 2018 and aggregates the hourly data to daily frequency. City-level weather records are first deduplicated at the city-hour level before daily summaries are computed. The energy variables are national daily indicators for Spain and therefore repeat across cities on the same date, while the weather variables vary by city.

Source

Nicholas J. H., "Hourly Energy Demand Generation and Weather", Kaggle. https://www.kaggle.com/datasets/nicholasjhana/energy-consumption-generation-prices-and-weather/data

According to the dataset description, electricity demand and generation data were obtained from ENTSOE, electricity prices from Red Electrica de Espana (REE), and weather data from the OpenWeather API for the five largest cities in Spain.

Examples

```
data(EnergyWx)
head(EnergyWx)

plot(load ~ temp,
     data = subset(EnergyWx, city == "Madrid"),
     xlab = "Daily mean temperature (C)",
     ylab = "Daily mean load")
```

FANG *Daily Prices for Four FANG Stocks*

Description

Daily stock prices for four FANG stocks redistributed from the **timetk** package and reorganized for use in **tisai**.

Usage

```
data(FANG)
```

Format

A data frame with 4032 observations on the following 8 variables:

`symbol` Stock ticker symbol: `AMZN`, `FB`, `GOOG`, or `NFLX`.

`date` Trading date from January 2, 2013 to December 30, 2016.

`open` Opening stock price in U.S. dollars.

`high` Highest stock price of the trading day in U.S. dollars.

`low` Lowest stock price of the trading day in U.S. dollars.

`close` Closing stock price in U.S. dollars.

`volume` Number of shares traded.

`adjusted` Adjusted closing stock price in U.S. dollars.

Details

The **tisai** version keeps the full sample provided by **timetk** and stores it as a base data frame that is ready for multi-asset time-series exercises, return calculations, and panel-style stock analyses.

Source

Redistributed from the **timetk** package. https://cran.r-project.org/package=timetk https://cran.r-project.org/web/packages/timetk/timetk.pdf

Examples

```
data(FANG)
head(FANG)

subset(FANG, symbol == "AMZN")[1:6, ]
```

`gafa_stock` *Daily Prices for Four GAFA Stocks*

Description

Daily stock prices for four GAFA stocks redistributed from the **fpp3** package, where the dataset is made available through **tsibbledata**, and reorganized for use in **tisai**.

Usage

```
data(gafa_stock)
```

Format

A data frame with 5032 observations on the following 8 variables:

`symbol` Stock ticker symbol: `AAPL`, `AMZN`, `FB`, or `GOOG`.

`date` Trading date from January 2, 2014 to December 31, 2018.

`open` Opening stock price in U.S. dollars.

`high` Highest stock price of the trading day in U.S. dollars.

`low` Lowest stock price of the trading day in U.S. dollars.

`close` Closing stock price in U.S. dollars.

`volume` Number of shares traded.

`adjusted` Adjusted closing stock price in U.S. dollars.

Details

In the **fpp3** ecosystem, `gafa_stock` is distributed as a `tsibble` on irregular trading days. The **tisai** version keeps the full sample and reorganizes it as a base data frame so it can be used directly in teaching examples on multi-asset prices, returns, and panel-style financial analysis.

Source

Available via the **fpp3** package, with the dataset object provided by **tsibbledata**. https://cran.r-project.org/package=fpp3 https://cran.r-project.org/web/packages/fpp3/fpp3.pdf https://cran.r-project.org/package=tsibbledata

The original **tsibbledata** documentation states that the underlying source is Yahoo Finance historical data.

Examples

```
data(gafa_stock)
head(gafa_stock)

subset(gafa_stock, symbol == "AAPL")[1:6, ]
```

get_Ashare_data — *Download A-share panel data from Yahoo Finance*

Description

This function downloads adjusted daily closing prices for selected A-share stocks from Yahoo Finance and constructs a long-format panel data set with daily log returns.

Usage

```
get_Ashare_data(
  tickers = NULL,
  from = "2018-01-01",
  to = "2025-12-31",
  save_path = NULL,
  ticker_info = NULL
)
```

Arguments

`tickers`	A character vector of Yahoo Finance ticker symbols, such as `c("600000.SS", "600036.SS", "000001.SZ")`. If `NULL`, the function will try to use the built-in ticker table.
`from`	Start date in `"YYYY-MM-DD"` format. Default is `"2018-01-01"`.
`to`	End date in `"YYYY-MM-DD"` format. Default is `"2025-12-31"`.
`save_path`	Optional file path for saving the resulting object as an `.rda` file. Default is `NULL`.
`ticker_info`	Optional data frame containing ticker information. It should include at least a column named `ticker`, and may also include `name` and `market`.

Details

If `tickers` is not provided, the function tries to use the built-in `Ashare` object (or `Ashare_tickers`, if available). Downloaded series are returned in a single long-format data frame sorted by ticker and date.

Value

A long-format data frame with columns:

date Trading date.

ticker Yahoo Finance ticker symbol.

name Company name, if available.

market Exchange code (`"SSE"` or `"SZSE"`), if available.

close_adj Adjusted closing price.

ret Daily log return.

Examples

```
## Not run:
data(Ashare)

panel <- get_Ashare_data(
  tickers = Ashare$ticker[1:5],
  from = "2020-01-01",
  to = "2024-12-31"
)

head(panel)

## End(Not run)
```

---

get_CNmarket_data — *Download Chinese financial market data from Yahoo Finance*

Description

This function downloads daily adjusted closing prices for major Chinese financial market indices and related financial variables from Yahoo Finance, aligns trading dates across markets, and returns a data frame.

Usage

```
get_CNmarket_data(from = "2018-01-01", to = "2025-12-31", save_path = NULL)
```

Arguments

`from`	A character string specifying the start date in `"YYYY-MM-DD"` format. The default is `"2018-01-01"`.
`to`	A character string specifying the end date in `"YYYY-MM-DD"` format. The default is `"2025-12-31"`.
`save_path`	An optional character string giving the file path to save the resulting data object as an `.rda` file. If `NULL`, the data are not saved to disk.

Details

The default series include the Shanghai Composite Index, Shenzhen Component Index, CSI 300 Index, Hang Seng Index, and the USD/CNY exchange rate.

The function retrieves market data from Yahoo Finance via `quantmod::getSymbols()`. Since different markets may have different trading calendars, the downloaded series are aligned by date using inner joins, so that only common trading days are retained.

The default symbols are:

shanghai Shanghai Composite Index (`000001.SS`)

shenzhen Shenzhen Component Index (`399001.SZ`)

hs300 CSI 300 Index (`000300.SS`)

hsi Hang Seng Index (`^HSI`)

usd_cny USD/CNY exchange rate (`CNY=X`)

Value

A data frame containing aligned daily adjusted closing prices for the selected Chinese and related financial market series. The first column is `date`, followed by the downloaded market variables.

Examples

```
## Not run:
library(tisai)

CNmarket <- get_CNmarket_data()
head(CNmarket)

CNmarket2 <- get_CNmarket_data(
  from = "2020-01-01",
  to = "2024-12-31"
)

CNmarket3 <- get_CNmarket_data(
  save_path = "CNmarket.rda"
)

## End(Not run)
```

`get_USmarket_data` *Download global financial market data from Yahoo Finance*

Description

This function downloads daily adjusted closing prices for major global financial market assets from Yahoo Finance, aligns trading dates across markets, and returns a data frame.

Usage

```
get_USmarket_data(
  from = "2018-01-01",
  to = "2025-12-31",
  symbols = NULL,
  save_path = NULL
)
```

Arguments

`from`	A character string specifying the start date in `"YYYY-MM-DD"` format. The default is `"2018-01-01"`.
`to`	A character string specifying the end date in `"YYYY-MM-DD"` format. The default is `"2025-12-31"`.
`symbols`	An optional named character vector of Yahoo Finance symbols. The names of this vector will be used as the output variable names. If `NULL`, a default set of symbols is used.
`save_path`	An optional character string giving the file path to save the resulting data object as an `.rda` file. If `NULL`, the data are not saved to disk.

Details

The default series include the S and P 500 index, Nasdaq Composite index, U.S. dollar index, and gold futures price.

The function retrieves market data from Yahoo Finance via `quantmod::getSymbols()`. Since different markets may have different trading calendars, the downloaded series are aligned by date using inner joins, so that only common trading days are retained.

The default symbols are:

sp500 S and P 500 Index (`^GSPC`)

nasdaq Nasdaq Composite Index (`^IXIC`)

dxy U.S. Dollar Index (`DX-Y.NYB`)

gold Gold Futures (`GC=F`)

Value

A data frame containing aligned daily adjusted closing prices for the selected market series. The first column is `date`, followed by the downloaded market variables.

Examples

```
## Not run:
library(tisai)

USmarket <- get_USmarket_data()
head(USmarket)

USmarket2 <- get_USmarket_data(
  from = "2020-01-01",
  to = "2024-12-31"
)

USmarket3 <- get_USmarket_data(
  save_path = "USmarket.rda"
)

## End(Not run)
```

GoldUSD — *Daily Gold Price Series in U.S. Dollars*

Description

Daily open, high, low, close, and trading volume data for the Gold/USD market from 2016 to 2026.

Usage

```
data(GoldUSD)
```

Format

A data frame with 2554 observations on the following 6 variables:

`date` Trading date from January 4, 2016 to March 3, 2026.

`open` Opening price in U.S. dollars.

`high` Highest price of the day in U.S. dollars.

`low` Lowest price of the day in U.S. dollars.

`close` Closing price in U.S. dollars.

`volume` Trading volume.

Details

The raw file `GoldUSD.csv` contains 6399 daily observations from August 30, 2000 to March 3, 2026. Following the user-approved trimming rule for this project, the package version keeps the observations from 2016 onward, which yields 2554 trading-day records. The sample starts on January 4, 2016 because that is the first trading day on or after January 1, 2016.

Source

Krupal Patel, "Gold Price Dynamics", Kaggle. https://www.kaggle.com/datasets/krupalpatel07/gold-price-dynamics

Examples

```
data(GoldUSD)
head(GoldUSD)

plot(close ~ date, data = GoldUSD, type = "l",
     xlab = "Date", ylab = "USD",
     main = "Gold/USD Closing Price")
```

`Gtemp` *Global Annual Temperature Anomalies (1850–2017)*

Description

This dataset provides the global annual mean temperature anomalies relative to the 1961–1990 average, sourced from the HadCRUT4 dataset.

Usage

```
data(Gtemp)
```

Format

A data frame with 168 observations on the following 2 variables:

`year` The year of observation (from 1850 to 2017).

`anomaly` Global temperature anomaly (degrees Celsius) relative to the 1961–1990 mean.

Details

The dataset reflects the long-term warming trend and potential 60-70 year periodic oscillations in global temperatures.

Source

Climatic Research Unit (CRU), University of East Anglia (HadCRUT4). https://crudata.uea.ac.uk/cru/data/temperature/

References

Wang, S., You, J., & Huang, T. (2022). Modelling and applications for non-stationary time series in the presence of trend and period. Scientia Sinica Mathematica, 52(2), 177-208.

Examples

```
data(Gtemp)
plot(Gtemp$year, Gtemp$anomaly, type = "l", main = "Global Temperature Anomalies")
```

---

gtemp.month *Global Temperature Monthly Data*

Description

Global temperature monthly data from the book "AI enabled time series and spacial statistics"

Usage

```
data(gtemp.month)
```

Format

A data frame with 12 rows (months) and 49 columns (years from 1975 to 2023). Each cell contains the average temperature for that month and year.

Details

Global temperature monthly data

This dataset contains global temperature monthly data from the book "AI enabled time series and spacial statistics".

The gtemp.month dataset provides monthly global temperature data from 1975 to 2023. Rows represent months (1-12) and columns represent years. This dataset is often used in climate change analysis and time series modeling.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(gtemp.month)

# Transpose the data for plotting
gtemp_t <- t(gtemp.month)

# Plot the temperature data for January
plot(rownames(gtemp_t), gtemp_t[, 1], type = "l",
     main = "January Global Temperature",
     xlab = "Year", ylab = "Temperature",
     col = "blue")

# Calculate summary statistics for each month
apply(gtemp.month, 1, summary)
```

`Hare` *Hare Population Data*

Description

Hare population data from the book "AI enabled time series and spacial statistics"

Usage

```
data(Hare)
```

Format

A time series object with the following characteristics:

- Time period: 1845 - 1935
- Frequency: Annual
- Values: Hare population counts

Details

Hare population data

This dataset contains hare population data from the book "AI enabled time series and spacial statistics".

The Hare dataset provides annual population counts of hares from 1845 to 1935. This dataset is often used in time series analysis and population dynamics studies.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(Hare)

# Plot the hare population data
plot(Hare, main = "Hare Population Data",
     xlab = "Year", ylab = "Population",
     col = "blue")

# Calculate summary statistics
summary(Hare)
```

HKflu *Weekly Influenza Surveillance and Environmental Data in Hong Kong*

Description

This dataset contains weekly records of influenza incidence rates along with various air quality and meteorological indicators in Hong Kong from December 30, 2012, to December 22, 2019.

Usage

```
data(HKflu)
```

Format

A data frame with 365 observations on the following 14 variables:

- `week` A numeric vector indicating the sequence of monitored weeks.
- `influenza` Influenza intensity, measured as the number of laboratory-confirmed influenza cases per 1,000 recorded consultations.
- `FSP` Fine Suspended Particulates (PM2.5) daily average concentration.
- `RSP` Respirable Suspended Particulates (PM10) daily average concentration.
- `NO2` Nitrogen Dioxide (NO_2) daily average concentration.
- `CO` Carbon Monoxide (CO) daily average concentration.
- `SO2` Sulphur Dioxide (SO_2) daily average concentration.
- `NOX` Nitrogen Oxides (NO_x) daily average concentration.
- `O3` Ozone (O_3) daily average concentration.
- `temperature` Weekly average temperature in degrees Celsius.
- `humidity` Weekly average relative humidity percentage.
- `start` Start date of the surveillance week.
- `end` End date of the surveillance week.
- `wee` Week index retained from the original source file.

Source

Hong Kong Environmental Protection Department and relevant health authorities.

References

Li, J., Wang, S., & You, J. (2022). Nonparametric additive models for discrete-time series with periodic features. Acta Mathematica Sinica, Chinese Series, 65(1), 177-204.

Examples

```
data(HKflu)
summary(HKflu)
plot(HKflu$temperature, HKflu$influenza)
```

HousePrice *Monthly House Price Index for Selected U.S. Cities*

Description

Monthly Zillow Home Value Index (ZHVI) for selected major U.S. cities.

Usage

```
data(HousePrice)
```

Format

A data frame containing:

`region` City name.

`state` State name.

`date` Month.

`price` Typical home value (USD).

Source

Zillow Research Housing Data. https://www.zillow.com/research/data/

---

lap *Local Area Pollution (LAP) Data*

Description

LAP data from the book "AI enabled time series and spacial statistics"

Usage

```
data(lap)
```

Format

A multivariate time series (mts) object with the following characteristics:

- Time period: 1970 - 1980
- Frequency: Weekly (52 observations per year)
- Number of variables: 11
- Variables:
  - tmort: Total mortality
  - rmort: Respiratory mortality
  - cmort: Cardiovascular mortality
  - tempr: Temperature
  - rh: Relative humidity
  - co: Carbon monoxide
  - so2: Sulfur dioxide
  - no2: Nitrogen dioxide
  - hycarb: Hydrocarbons
  - o3: Ozone
  - part: Particulate matter

Details

LAP data

This dataset contains LAP (Local Area Pollution) data from the book "AI enabled time series and spacial statistics".

The lap dataset provides weekly measurements of various pollution and health indicators from 1970 to 1980. This dataset is often used in environmental health studies and time series analysis of pollution effects.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(lap)

# Plot the first variable (total mortality)
plot(lap[, 1], main = "Total Mortality",
     xlab = "Time", ylab = "Mortality Rate",
     col = "red")

# Calculate correlation between temperature and ozone
cor(lap[, "tempr"], lap[, "o3"], use = "complete.obs")

# Plot multiple variables
plot(lap[, c("tempr", "o3", "co")],
     main = "Temperature, Ozone, and Carbon Monoxide")
```

Lynx *Lynx Population Data*

Description

Lynx population data from the book "AI enabled time series and spacial statistics"

Usage

```
data(Lynx)
```

Format

A time series object with the following characteristics:

- Time period: 1845 - 1935
- Frequency: Annual
- Values: Lynx population counts

Details

Lynx population data

This dataset contains lynx population data from the book "AI enabled time series and spacial statistics".

The Lynx dataset provides annual population counts of lynx from 1845 to 1935. This dataset is often used in time series analysis and predator-prey relationship studies.

Source

Package 'astsa' available at https://nickpoison.github.io/

Examples

```
data(Lynx)

# Plot the lynx population data
plot(Lynx, main = "Lynx Population Data",
     xlab = "Year", ylab = "Population",
     col = "red")

# Calculate summary statistics
summary(Lynx)
```

`poverty` *Global Poverty, Inequality, and GDP Panel Data*

Description

Country-year panel data on population, GDP, poverty, and income distribution indicators from 2005 to 2024.

Usage

```
data(poverty)
```

Format

A data frame with 3980 observations on the following 11 variables:

`country` Country or territory name.

`iso_code` Three-letter country or territory code.

`year` Calendar year from 2005 to 2024.

`pop` Population.

`gdp` Gross domestic product.

`gdppc` GDP per capita.

`poverty` Poverty rate.

`gini` Gini index.

`top1` Income share of the top 1 percent.

`top10` Income share of the top 10 percent.

`bottom50` Income share of the bottom 50 percent.

Details

This dataset is drawn from the Kaggle collection "Global Economic Inequality and Poverty 1980–2024". Following the user-approved trimming rule for this project, the package version keeps the latest full years that fit within the 4000-row limit, which currently yields a country-year panel from 2005 to 2024 with 3980 observations for 199 countries and territories.

Missing values from the source are preserved rather than imputed. In the current package version, `gdp` and `gdppc` each contain 1028 missing values, `poverty` and `gini` each contain 2507 missing values, and `top1`, `top10`, and `bottom50` each contain 300 missing values.

Source

Luca Lullo, "Global Economic Inequality and Poverty 1980–2024", Kaggle. https://www.kaggle.com/datasets/lucalullo/global-economic-inequality-and-poverty-1980-2024

The dataset description reports underlying data sources including Our World in Data and the World Inequality Database (WID), with poverty and inequality indicators mainly based on World Bank household survey data. https://ourworldindata.org/ https://wid.world/

Examples

```
data(poverty)
head(poverty)

plot(gdppc ~ year, data = subset(poverty, country == "China"),
     type = "b", xlab = "Year", ylab = "GDP per capita",
     main = "China GDP per Capita")
```

Power *Hourly Household Electricity Consumption*

Description

Hourly electricity consumption of a household in 2010.

Usage

```
data(Power)
```

Format

A data frame containing:

`time` Hourly timestamp

`power` Active power (kilowatt)

`reactive` Reactive power (kilowatt)

`voltage` Voltage (volt)

`current` Current intensity (ampere)

`sub1` Energy sub-metering 1

`sub2` Energy sub-metering 2

`sub3` Energy sub-metering 3

Source

UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/individual+household+electric+power+consumption `retail` *Monthly U.S. Retail Sales, Inventories, and Selected Store Categories*

Description

Monthly U.S. retail indicators from the U.S. Census Bureau, including aggregate retail sales, inventories, inventory-sales ratio, and selected seasonally adjusted store-category sales.

Usage

```
data(retail)
```

Format

A data frame with 332 observations on the following 13 variables:

- `date` Month of observation from February 1992 to September 2019.
- `salesn` Retail sales excluding food services, not seasonally adjusted, in millions of dollars.
- `sales` Retail sales excluding food services, seasonally adjusted, in millions of dollars.
- `growthn` Monthly percent change in retail sales, not seasonally adjusted.
- `growth` Monthly percent change in retail sales, seasonally adjusted.
- `total` Retail and food services total sales, seasonally adjusted, in millions of dollars.
- `invent` Retailers inventories, seasonally adjusted, in millions of dollars.
- `ratio` Retailers inventory-to-sales ratio, seasonally adjusted.
- `auto` Automotive parts, accessories, and tire stores sales, seasonally adjusted, in millions of dollars.
- `furn` Furniture and home furnishings stores sales, seasonally adjusted, in millions of dollars.
- `build` Building materials, garden equipment, and supplies dealers sales, seasonally adjusted, in millions of dollars.
- `cloth` Clothing and clothing accessory stores sales, seasonally adjusted, in millions of dollars.
- `online` Electronic shopping and mail-order houses sales, seasonally adjusted, in millions of dollars.

Details

The raw directory contains multiple monthly series downloaded from a Kaggle collection that republishes U.S. Census Bureau retail data distributed through FRED. To build a compact teaching dataset, the series are merged into a single monthly data frame covering February 1992 to September 2019. The sample starts in February 1992 because the month-over-month percent change series begin one month later than the level series.

Source

U.S. Census Bureau, Monthly Retail Trade Survey, distributed by the Federal Reserve Bank of St. Louis (FRED). https://fred.stlouisfed.org/ https://www.census.gov/retail/index.html

Raw files in this project were collected from the Kaggle dataset “Retail and Retailers Sales Time Series Collection”. https://www.kaggle.com/datasets/census/retail-and-retailers-sales-time-series-coll

Examples

```
data(retail)
head(retail)

plot(retail$date, retail$sales, type = "l",
     xlab = "Date", ylab = "Millions of dollars",
     main = "U.S. Retail Sales")
```

`SCarinae` *Ten-Day Visual Magnitude Series for S Carinae*

Description

Ten-day visual magnitude series for the variable star S Carinae, reorganized for use in **tisai**.

Usage

```
data(SCarinae)
```

Format

A data frame with 1189 observations on the following 2 variables:

`jd` Julian day index from 2431110 to 2442990 in 10-day steps.

`magnitude` Visual magnitude of S Carinae. Missing 10-day intervals are recorded as `NA`. Smaller values indicate a brighter star.

Details

The underlying observations are visual measurements of the southern variable star S Carinae. According to the project notes, the usable sample spans JD2431110 to JD2442990, corresponding to January 21, 1944 through July 30, 1976. The released **tisai** object uses the provided 10-day series and converts placeholder zeros to `NA`; there are 40 such missing intervals in the current sample.

Source

The official observational source is AAVSO WebObs for S Carinae. https://apps.aavso.org/webobs/results/?star=000-BBR-212&num_results=200&page=158

Project notes also reference the time-series data page maintained by Guadalupe Huerta: https://math.unm.edu/~ghuerta/tseries/data.html

The released **tisai** version is built from the local processed 10-day series file supplied in this project.

Examples

```
data(SCarinae)
head(SCarinae)

plot(SCarinae$jd, SCarinae$magnitude, type = "l",
     xlab = "Julian day", ylab = "Magnitude",
     main = "S Carinae")
```

sunspots — *Monthly Sunspot Numbers*

Description

Monthly mean relative sunspot numbers redistributed from the base R **datasets** package.

Usage

```
data(sunspots)
```

Format

A univariate time series with 2820 monthly observations from January 1749 to December 1983.

Details

This is the classical monthly sunspot series used in many forecasting and spectral-analysis examples. The object is kept as a `ts` series in **tisai** so it can be used directly in time-series demonstrations.

Source

Redistributed from the base R **datasets** package.

Andrews, D. F. and Herzberg, A. M. (1985). *Data: A Collection of Problems from Many Fields for the Student and Research Worker*. Springer.

The **datasets** documentation notes that the figures were collected at the Swiss Federal Observatory, Zurich until 1960, and then at the Tokyo Astronomical Observatory.

Examples

```
data(sunspots)
plot(sunspots, main = "Monthly Sunspot Numbers")
```

TempAnom — *Global Surface Air Temperature Anomalies from Hansen and Lebedeff*

Description

Annual global surface air temperature anomalies extracted from Hansen and Lebedeff (1987) and reorganized for use in **tisai**.

Usage

```
data(TempAnom)
```

Format

A data frame with 106 observations on the following 2 variables:

`year` Year of observation from 1880 to 1985.

`anomaly` Global surface air temperature anomaly in degrees Celsius relative to the 1951–1980 mean.

Details

This series corresponds to the annual global temperature anomaly values discussed in Shao (2011) and traced through Hall and Van Keilegom (2003) back to Table 1 of Hansen and Lebedeff (1987). It provides an early annual global temperature anomaly series based on station observations with a different reference period from the `Gtemp` dataset in **tisai**.

Source

Hansen, J. and Lebedeff, S. (1987). Global trends of measured surface air temperature. *Journal of Geophysical Research: Atmospheres*, 92(D11), 13345–13372.

The file in this project was prepared from the table cited through:

Hall, P. and Van Keilegom, I. (2003). Using difference-based methods for inference in nonparametric regression with time series errors. *Journal of the Royal Statistical Society: Series B*, 65(2), 443–456.

Shao, X. (2011). A simple test of changes in mean in the possible presence of long-range dependence. *Journal of Time Series Analysis*, 32(6), 598–606.

Examples

```
data(TempAnom)
head(TempAnom)

plot(TempAnom$year, TempAnom$anomaly, type = "l",
     xlab = "Year", ylab = "Temperature anomaly",
     main = "Hansen-Lebedeff Global Temperature Anomalies")
```

`USeconomic` *U.S. Quarterly National Economic Output and Expenditure Components*

Description

This dataset provides a structural (quarterly) snapshot of the U.S. economy from 2000.1.1 to 2025.7.1, consisting of 103 observations.

It integrates the major components of Gross Domestic Product (GDP): - Aggregate Output: Nominal and Real GDP. - Domestic Demand: Personal consumption (PCE) and Private investment. - External Sector: Real exports and imports of goods and services.

The dataset allows students to explore the nonlinear dynamics of investment shocks and the global trade balance using advanced AI structural models.

Usage

```
data(USeconomic)
```

Format

A data frame with aligned quarterly observations on the following 7 variables:

`date` Start date of the quarter (YYYY-MM-DD).

`gdp` Gross Domestic Product (Billions of Dollars).

`real_gdp` Real Gross Domestic Product (Billions of Chained 2017 Dollars).

`pce` Personal Consumption Expenditures (Billions of Dollars).

`invest` Real Gross Private Domestic Investment (Billions of Chained 2017 Dollars).

`export` Real Exports of Goods and Services (Billions of Chained 2017 Dollars).

`import` Real Imports of Goods and Services (Billions of Chained 2017 Dollars).

Source

U.S. Bureau of Economic Analysis (BEA) via Federal Reserve Bank of St. Louis (FRED). Data IDs: GDP, GDPC1, PCE, GPDIC1, EXPGSC1, and IMPGSC1. Visit: https://fred.stlouisfed.org/

References

U.S. Bureau of Economic Analysis, "National Income and Product Accounts". Retrieved from FRED: https://fred.stlouisfed.org/categories/106

Examples

```
data(USeconomic)
# Calculate Investment-to-GDP ratio
inv_ratio <- USeconomic$invest / USeconomic$real_gdp
plot(USeconomic$date, inv_ratio, type="l", ylab="Investment/GDP Ratio")
```

`USmacro` *U.S. Monthly Macroeconomic Pulse: Labor, Price, and Monetary Indicators*

Description

This dataset provides a high-frequency (monthly) snapshot of the U.S. economy from 2000.1.1 to 2025.10.1, consisting of 310 observations.

The variables cover four critical dimensions: 1. Labor Market: Unemployment rate and Nonfarm payrolls. 2. Price Level: Consumer Price Index (CPI) for inflation tracking. 3. Monetary Policy: Federal Funds Rate and 10-Year Treasury yields. 4. Real Activity: Industrial production and Housing starts.

This data is ideal for teaching lead-lag relationships, regime switching, and high-frequency forecasting using LSTM or Transformer architectures.

Usage

```
data(USmacro)
```

Format

A data frame with aligned monthly observations on the following 8 variables:

`date` First day of the month (YYYY-MM-DD).

`unrate` Civilian Unemployment Rate (Percent).

`cpi` Consumer Price Index for All Urban Consumers (Index 1982-1984=100).

`interest` Federal Funds Effective Rate (Percent).

`indpro` Industrial Production Index (Index 2017=100).

`employ` All Employees, Total Nonfarm (Thousands of Persons).

`housing` New Privately Owned Housing Units Started (Thousands of Units).

`bond10y` 10-Year Treasury Constant Maturity Rate (Percent).

Source

Federal Reserve Bank of St. Louis (FRED). Data accessible via the following IDs: UNRATE, CPIAUCSL, FEDFUNDS, INDPRO, PAYEMS, HOUST, and GS10. Visit: https://fred.stlouisfed.org/

References

Federal Reserve Bank of St. Louis, Federal Reserve Economic Data (FRED). https://fred.stlouisfed.org

Examples

```
data(USmacro)
# Visualize the relationship between interest rates and housing starts
plot(USmacro$interest, USmacro$housing,
     main="Interest Rates vs Housing Starts")
```

`walmart_sales_weekly` *Weekly Walmart Sales Sample*

Description

Weekly Walmart sales data redistributed from the **timetk** package and reorganized for use in **tisai**.

Usage

```
data(walmart_sales_weekly)
```

Format

A data frame with 1001 observations on the following 17 variables:

`id` Series identifier combining store and department.

`store` Store identifier.

`dept` Department identifier.

`date` Weekly date from February 5, 2010 to October 26, 2012.

`weekly_sales` Weekly department sales.

`is_holiday` Logical indicator for holiday weeks.

`type` Store type.

`size` Store size in square feet.

`temperature` Regional average temperature.

`fuel_price` Regional fuel price.

`markdown1` Promotional markdown feature 1.

`markdown2` Promotional markdown feature 2.

`markdown3` Promotional markdown feature 3.

`markdown4` Promotional markdown feature 4.

`markdown5` Promotional markdown feature 5.

`cpi` Consumer price index.

`unemployment` Regional unemployment rate.

Details

The original **timetk** dataset is a sample derived from the Kaggle competition “Walmart Recruiting - Store Sales Forecasting”. The **tisai** version keeps the same sample coverage, which contains seven department-level weekly series from Store 1, while standardizing the object as a base data frame with lowercase variable names. Missing promotional markdown values are preserved.

Source

Redistributed from the **timetk** package. https://cran.r-project.org/package=timetk https://cran.r-project.org/web/packages/timetk/timetk.pdf

In **timetk**, this dataset is described as a sample derived from the Kaggle competition “Walmart Recruiting - Store Sales Forecasting”. https://www.kaggle.com/c/walmart-recruiting-store-sales-forecasting

Examples

```
data(walmart_sales_weekly)
head(walmart_sales_weekly)

subset(walmart_sales_weekly, dept == 1)[1:6, ]
```

`WorldMacro` *Global Macroeconomic Panel Data*

Description

Panel data of macroeconomic indicators for multiple countries, including GDP per capita, population, and inflation rate.

Usage

```
data(WorldMacro)
```

Format

A data frame containing:

`country` Country name.

`iso2c` Two-letter country code.

`iso3c` Three-letter country code.

`year` Year.

`gdp_pc` GDP per capita (constant USD).

`population` Total population.

`inflation` Inflation rate.

Source

World Bank World Development Indicators https://data.worldbank.org

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